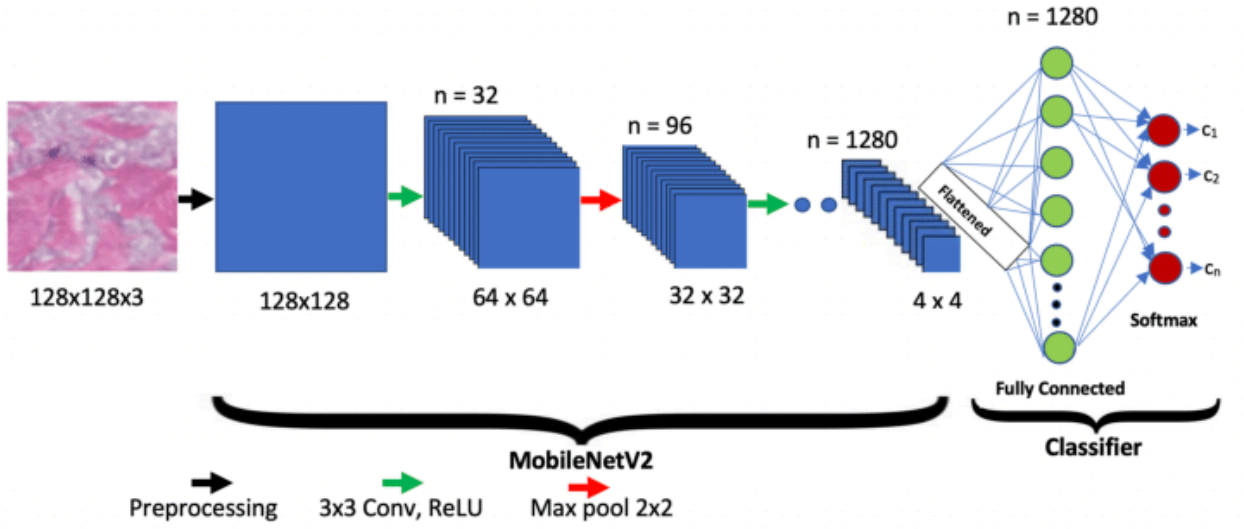
Mobilenetv2 Architecture:
MobileNetV2 is a convolutional neural network architecture that seeks to perform well on mobile devices. It is based on an inverted residual structure where the residual connections are between the bottleneck layers. The intermediate expansion layer uses lightweight depthwise convolutions to filter features as a source of non-linearity. As a whole, the architecture of MobileNetV2 contains the initial fully convolution layer with 32 filters, followed by 19 residual bottleneck layers.



MobileNetV2 is designed for **speed and efficiency** (great for Raspberry Pi & mobile). It introduces two major ideas:

1. **Depthwise Separable Convolutions** – Makes model lightweight.
2. **Inverted Residual Blocks** – Instead of reducing dimensions then expanding, it expands then compresses.

Stage	Input Resolution	Operator	t (expansion)	c (output channels)	n (repeats)	s (stride)
0	224×224×3	Conv2D 3×3	-	32	1	2
1	112×112×32	Bottleneck (3×3 dw)	1	16	1	1
2	112×112×16	Bottleneck (3×3 dw)	6	24	2	2

3	56×56×24	Bottleneck (3×3 dw)	6	32	3	2
4	28×28×32	Bottleneck (3×3 dw)	6	64	4	2
5	14×14×64	Bottleneck (3×3 dw)	6	96	3	1
6	14×14×96	Bottleneck (3×3 dw)	6	160	3	2
7	7×7×160	Bottleneck (3×3 dw)	6	320	1	1
8	7×7×320	Conv2D 1×1	-	1280	1	1
9	7×7×1280	Global Average Pooling	-	-	-	-
10	1×1×1280	Fully Connected (Dense)	-	num_classes	1	-

- Input goes from **high resolution + few channels**
- To **low resolution + many channels**
- MobileNetV2 is very efficient because it:
    - Avoids heavy convolution
    - Uses shortcut (residual) connections smartly
    - Uses depthwise separable

MobileNetV2 Architecture Summary (Explained)

Stage	Input Resolution	Operator	t (expansion)	c (output channels)	n (repeats)	s (stride)
0	224×224×3	Conv2D 3×3	-	32	1	2

🔹 **Stage 0:**

- Input image: size 224×224, 3 channels (RGB)
- Apply normal 3×3 convolution
- Output has 32 feature maps
- **Stride = 2**, so size reduces to 112×112×32

`For this:`

- The network starts reducing image size early to save computation.
- It increases depth (channels) while shrinking width and height.
- Sets up a strong foundation for deeper layers to extract more abstract features.

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